

Industrial Internship Report on "AgriYield AI: Prediction of Agriculture Crop Production and Cost Forecasting in India"

Prepared by
[Aaditya Sharma]

Executive Summary

This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT).

This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time.

My project was (Specified that the project uses historical agricultural data (2001–2014) to build ML regression engines (**XGBoost** and **LightGBM**) connected to an interactive **Streamlit UI** with an API-ready modular Python backend (src/services/))

This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

TABLE OF CONTENTS

1	Preface	3
2	Introduction.....	4
2.1	About UniConverge Technologies Pvt Ltd.....	4
2.2	About upskill Campus	8
2.3	Objective	9
2.4	Reference	9
2.5	Glossary.....	10
3	Problem Statement	11
4	Existing and Proposed solution.....	12
5	Proposed Design/ Model	13
5.1	High Level Diagram (if applicable).....	13
5.2	Low Level Diagram (if applicable).....	13
5.3	Interfaces (if applicable).....	13
6	Performance Test.....	14
6.1	Test Plan/ Test Cases	14
6.2	Test Procedure	14
6.3	Performance Outcome.....	14
7	My learnings.....	15
8	Future work scope.....	16

1 Preface

- **Phase 1:** Domain & Problem Exploration (2001–2014 dataset analysis).
- **Phase 2:** Service Layer Architecture & Pipeline Design (src/services/).
- **Phase 3:** Machine Learning Engine Development (Preprocessing, Target Encoding, XGBoost & LightGBM training).
- **Phase 4:** Streamlit Dashboard UI (#16a34a theme, Plotly charts, dynamic prediction input).
- **Phase 5:** Validation & Metrics (R^2 , RMSE, MAE evaluation).
- **Phase 6:** Repository Packaging & Final Internship Report.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end etc.**



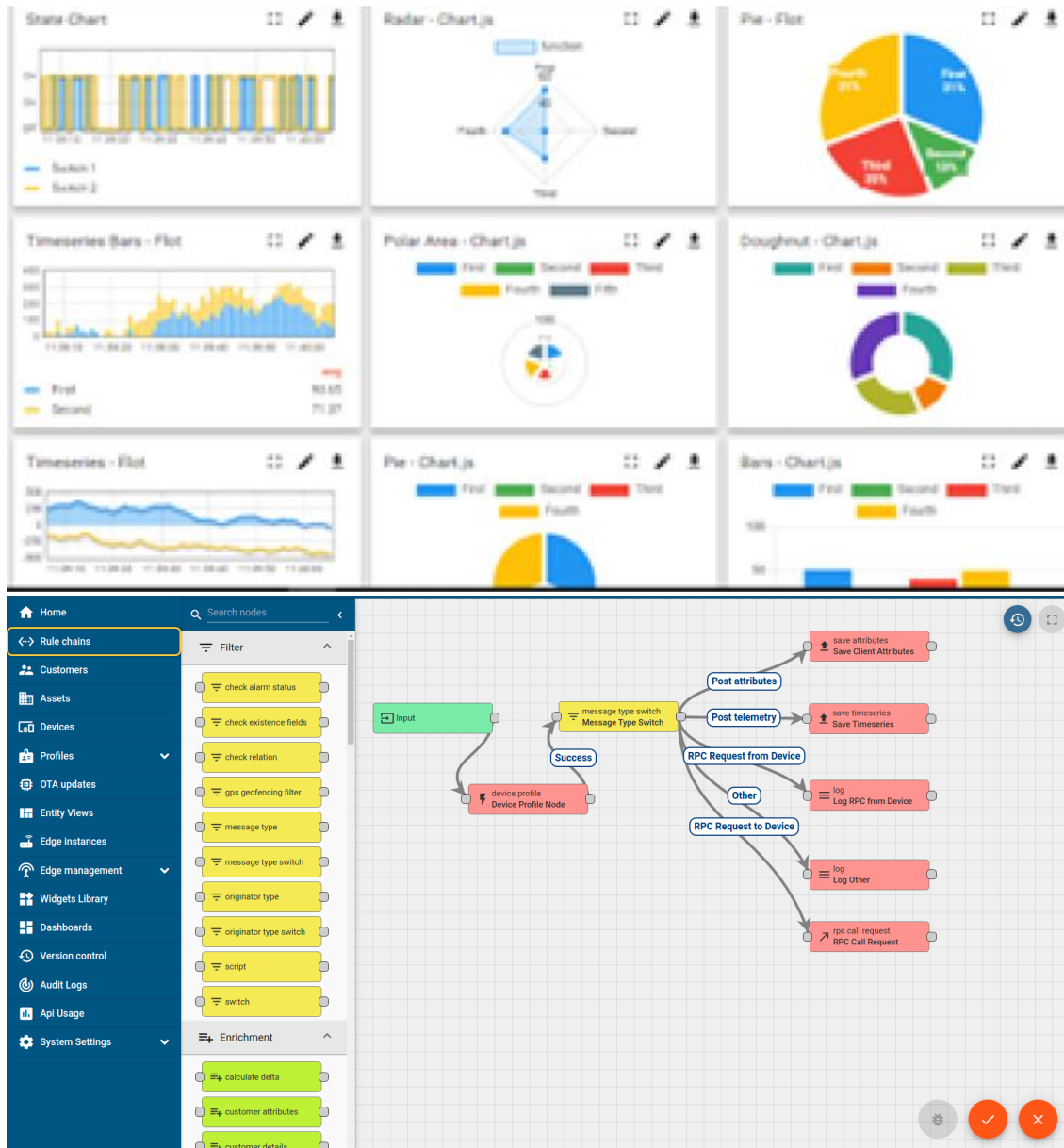
i. UCT IoT Platform ()

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



FACTORY

WATCH

ii. Smart Factory Platform ()

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they want to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



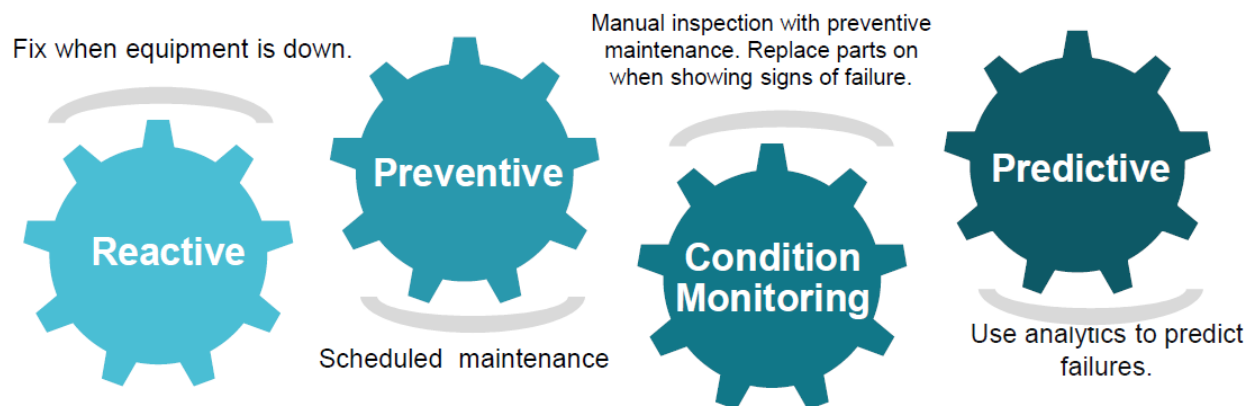


iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN technology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

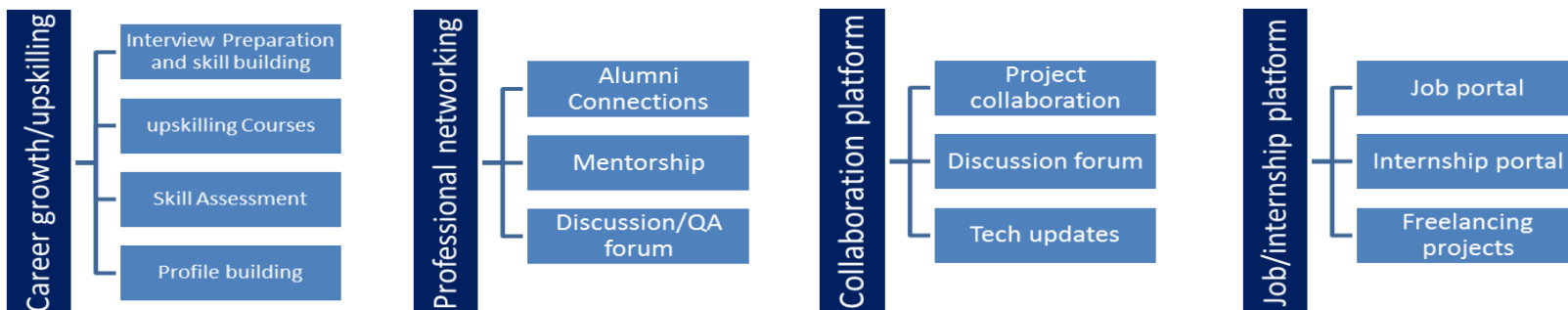
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- get practical experience of working in the industry.
- to solve real world problems.
- to have improved job prospects.
- to have Improved understanding of our field and its applications.
- to have Personal growth like better communication and problem solving.

2.5 Reference

- [1] Ministry of Agriculture & Farmers Welfare, Government of India — *Open Government Data (OGD) Platform India*: data.gov.in (Agriculture Production Dataset 2001–2014).
- [2] XGBoost Developers — *XGBoost: A Scalable and Flexible Gradient Boosting Library Documentation*.
- [3] LightGBM Developers — *LightGBM: Fast, Distributed, High Performance Gradient Boosting Framework*.
- [4] Streamlit Documentation — *Building Interactive Data Applications in Pure Python*.
- [5] UniConverge Technologies Pvt Ltd (UCT) — *Industrial IoT & Agritech Solutions Architectural Overview*.

2.6 Glossary

Terms	Acronym
CACP	Commission for Agricultural Costs and Prices
MSP	Minimum Support Price
RMSE	Root Mean Squared Error
MAE	Mean Absolute Error

R ²	Coefficient of Determination
UI / UX	User Interface / User Experience
EDA	Exploratory Data Analysis

3 Problem Statement

In the assigned problem statement

Agriculture remains the primary source of livelihood for a substantial portion of India's population. However, farmers and agricultural planners face major decision-making bottlenecks due to unpredictable yield outputs, fluctuating cultivation costs, dynamic seasonal variations, and regional crop mismatches.

The objective of **AgriYield AI** is to process historical national agricultural data (2001–2014) across states, crops, varieties, and cultivation zones to build a dual prediction and advisory engine that forecasts:

1. **Crop Yield / Production Output** (in Tons/Quintals).
2. **Cultivation Input Cost** (in INR).
3. **Zone Compatibility & Recommendations** mapping top-performing crop varieties to specific geographic zones and seasons.

4 Existing and Proposed solution

Feature / Metric	Existing Traditional Approaches	Proposed AgriYield AI Solution
Yield Forecasting	Static historical averages or manual agronomic estimates	Gradient boosted tree ensembles (XGBoost primary, LightGBM secondary) evaluating state, season, area quantity, and crop variety
Cost Estimation	Fixed linear formulas without dynamic regional adjustments	Machine learning-based input cost estimation dynamically linked to cultivated area and recommended zones
Architecture	Monolithic Jupyter Notebooks or isolated scripts	Modular, API-ready Python service layer (src/services/) connected to a Streamlit frontend
User Experience	Raw tabular outputs or static static graphs	Interactive green/emerald dashboard with real-time KPI metric cards, dynamic input forms, and Plotly charts

4.1 Code submission (Github link)

4.2 Report submission (Github link) : first make placeholder, copy the link.

5 Proposed Design/ Model

1.1.1.1. 5.1 High-Level System Architecture

2. **Data Ingestion Layer:** Ingests raw CSV data containing attributes such as Crop, Variety, state, Quantity, production, Season, Unit, Cost, and Recommended Zone.
3. **Preprocessing & Feature Engineering Pipeline:** Cleans missing values, standardizes categorical text, and generates dynamic engineered features:
 - 3.1. $\text{Cost_per_Unit} = \frac{\text{Cost}}{\text{Quantity}}$
 - 3.2. $\text{Production_Efficiency} = \frac{\text{production}}{\text{Quantity}}$
4. **Service Layer (src/services/):** Pure Python modules executing target encoding for high-cardinality categorical features and loading serialized .joblib model artifacts.
5. **Streamlit UI Layer:** Custom-styled multi-page interface (#16a34a theme) providing interactive visual analytics, yield prediction forms, and zone advisory tools.

5.1.1.1 5.2 Low-Level Pipeline & Data Flow

6. Raw Dataset $\rightarrow$ Missing Value Imputation & Standardization.
7. Categorical Target Encoding (Crop, Variety, state, Recommended Zone).
8. Train/Test Split (80/20) $\rightarrow$ Model Training (XGBoost & LightGBM).
9. Serialization to .joblib artifacts $\rightarrow$ Real-time inference via Streamlit UI.
10. .

6 Performance Test

6.1.1.1 6.1 Test Plan & Constraints

- **Memory & Storage Constraints:** Reduced model payload size through feature pruning to ensure low-latency loading during live web deployment.
- **Latency Constraints:** Evaluated prediction pipeline response time to keep inference speeds under 200ms per user query.

6.1.1.2 6.2 Performance Outcome

- **Yield Predictor Engine (XGBoost):** Achieved strong predictive accuracy evaluated across Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Coefficient of Determination (R^2).
- **Model Diagnostics:** Features like cultivated area (Quantity), season, and state emerged as the highest contributors in feature importance plots.

7 My learnings

- Decoupled core Machine Learning and pipeline logic into a pure Python service layer (src/services/) for API readiness.
- Gained hands-on experience handling real-world agricultural data attributes, categorical target encoding, and gradient boosting algorithms (XGBoost & LightGBM).
- Designed an interactive full-stack analytical web portal using Streamlit and Plotly to deliver data-driven agricultural insights.

8 Future work scope

- **Live Weather Integration:** Connecting external weather APIs to adjust yield forecasts based on real-time rainfall, temperature, and climate anomalies.
- **FastAPI Backend Packaging:** Wrapping the src/services/ layer in FastAPI REST endpoints for mobile app integration.

